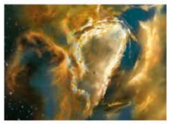
One of the brightest nebulae in the night sky, the Orion Nebula is a major star-forming region, lying at a distance of 1,340 light years. It is easily visible through binoculars in the night sky, although it will not be as bright as seen here.

HOW STARS FORM

Gravity plays a vital role in starbirth, pulling together the gas and dust in a galaxy until it condenses into a shining star.

1 Starbirth begins in a dense molecular cloud, rich in molecules such as hydrogen and carbon monoxide.



2 Gravity causes the cloud's centre to collapse into denser fragments, which will form a cluster of stars.

3 The shrinking gas in each fragment grows hot enough to emit infrared rays: it is now a "protostar".

4 Spinning gas and dust around the protostar form a thin disc: hot gas streams out from its poles.

5 The centre of the protostar becomes hot enough to ignite nuclear reactions: a star is born.

Three new stars are born in the Milky Way every year.